

[CTP-201] CCDB Counters versus raw ctp counters Created: 11/Jun/25 Updated: 26/Aug/25

Status:	Open
Project:	ALICE Central Trigger Processor
Component/s:	None
Affects Version/s:	None
Fix Version/s:	None

Type:	Task	Priority:	Minor
Reporter:	Roman Lietava	Assignee:	Jan Musinsky
Resolution:	Unresolved	Votes:	0
Labels:	None		
Remaining Estimate:	Not Specified		
Time Spent:	Not Specified		
Original Estimate:	Not Specified		

Change scope:	Manifest change
Checklist:	☒ EMPTY

Description

CTP Counters
All ctp counters are dumped to one file per day on tigger@ctpcptr (and in grafana).
Subset of all ctp counters relevant for lumi calculation is saved for every run in CCDB.
For introduction to class counters see:
[Counters1](#)
[Counters2](#)
Note that word scaler is synonym of counter.
Class counters have to satisfy 'causality conditions':

- scalers should never decrease
- LMB >= LMA >= LOB >= LOA >= L1B >= L1A: 5 relations
- clambIndex >= clamaIndex >= cla0bIndex >= cla0aIndex >= cla1bIndex >= cla1aIndex where Index is class index and relation is to be cheked for ALL classes in run.
- clamb1 >= clama1 >= cla0b1 >= cla0a1 >= cla1b1 >= cla1a1
The consistencyCeck is implemented in code at:
[o2::ctp::Scalers.cxx consistency check](#)
Errors are suppressed by if (eCnts.lmB < eCnts.MAXPRINT) { ... }

Steps in project:

- read run scalers from CCDB , see example in [CCDB macro](#)
- put eCnts.MAXPRINT to zero and find run with errors
- **follow the run by analysing ctp raw counters using your custom independent code and find bug on o2 scalers or confirm incosnstency in ctp raw counters**

Index in counter record can be found from vector at the end of file:
<https://github.com/AliceO2Group/AliceO2/blob/dev/DataFormats/Detectors/CTP/src/Scalers.cxx>
Class indexes [0..63] used in a run can be obtianed from CTPConfig in CCDB by auto clsindexes = ctpcfg->getTriggerClassList();